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Question 1: Linear Model (No Preferences)

Problem overview. The problem we aim to solve is a TA assignment and cost minimization problem. We want to decide how many BUs to assign to each TA applicant for each math course, such that all course BU needs are met, all applicant BU requests are satisfied, and the total cost (including prep BUs) is minimized.

Model: Parameters.

Applicants. There are 20 applicants, denoted $a_i \in \mathcal{A} = \{a_1, \dots, a_{20}\}$. Each applicant belongs to one of two programs:

- Ph.D. students, paid at wage $w_{\text{PhD}} = \$1,662$ per BU.
- MSc students, paid at wage $w_{\text{MSc}} = \$1,432$ per BU.

We write w_i for the wage per BU of applicant i , where $w_i = w_{\text{PhD}}$ or $w_i = w_{\text{MSc}}$ depending on their program.

Positions. There are 9 open positions, partitioned by appointment type:

$\mathcal{J} = \{\text{MATH150, MATH254, MACM316, MATH260, MATH338, MATH342, MATH426, MATHACW, MATHCW}\}$

$\mathcal{W} = \{\text{MATHACW, MATHCW}\}$ (workshops)

$\mathcal{T} = \{\text{MATH254, MACM316, MATH260, MATH342}\}$ (tutorials)

$\mathcal{S} = \{\text{MATH150}\}$ (seminars)

$\mathcal{M} = \{\text{MATH338, MATH426}\}$ (marking)

We also define $\mathcal{J}_{\text{TS}} = \mathcal{T} \cup \mathcal{S}$ as the set of tutorial and seminar positions.

BU parameters. Each appointment type has a BU step size s_j restricting how many BUs can be assigned:

- $s_j = \text{any real value}$, for workshops ($j \in \mathcal{W}$).
- $s_j = 0.5$, for marking and seminars ($j \in \mathcal{M} \cup \mathcal{S}$).
- $s_j = 1$, for tutorials ($j \in \mathcal{T}$).

Each position $j \in \mathcal{J}$ has a minimum and maximum total non-prep BU need:

$$b_j^{\min} \leq \text{total BUs at position } j \leq b_j^{\max}.$$

Course	Type	Min BUs ($b_j^{\min}$)	Max BUs ($b_j^{\max}$)
MATH150	SEM	1.5	2.0
MATH254	TUT	3.0	3.0
MACM316	TUT	9.0	9.0
MATH260	TUT	6.0	7.0
MATH338	MRK	0.5	1.5
MATH342	TUT	3.0	3.0
MATH426	MRK	0.5	1.5
MATHACW	WKP	20	25
MATHCW	WKP	12	16

Each applicant $i \in \mathcal{A}$ requests a minimum and maximum number of non-prep BUs:

$$\ell_i \leq \text{total non-prep BUs worked by } i \leq u_i.$$

On top of worked BUs, each applicant i is paid prep BUs p_{ij} for each position j they are assigned to. Prep BUs depend on both the applicant and the appointment type:

- p_{ij}^{WKP} : prep BUs for workshop appointments (varies by applicant).
- p_{ij}^{TUT} : prep BUs for tutorial appointments (varies by applicant).
- p_{ij}^{SEM} : prep BUs for seminar appointments (varies by applicant).
- $p_{ij}^{\text{MRK}} = 0$: no prep BUs for marking.

Prep BUs are paid once per position assigned and are not counted toward the applicant's BU request range. We write p_{ij} for the prep BUs of applicant i at position j , where the value is looked up from the data by appointment type.

Model: Decision Variables.

- $x_{ij} \geq 0$: non-prep BUs assigned to applicant i at position j . For workshops ($j \in \mathcal{W}$), x_{ij} is continuous. For tutorials ($j \in \mathcal{T}$), x_{ij} is a non-negative integer. For seminars and marking ($j \in \mathcal{M} \cup \mathcal{S}$), x_{ij} is a non-negative multiple of 0.5.
- $y_{ij} \in \{0, 1\}$: equals 1 if applicant i is assigned to position j , and 0 otherwise.

Model: Objective Function. Minimize total cost including both worked BUs and prep BUs, weighted by each applicant's wage:

$$\min \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i (x_{ij} + p_{ij} \cdot y_{ij}).$$

Constraints.

1. **Position BU needs are satisfied.** For each position $j \in \mathcal{J}$:

$$b_j^{\min} \leq \sum_{i \in \mathcal{A}} x_{ij} \leq b_j^{\max}.$$

2. **Applicant BU requests are satisfied.** For each applicant $i \in \mathcal{A}$:

$$\ell_i \leq \sum_{j \in \mathcal{J}} x_{ij} \leq u_i.$$

3. **BU step consistency.** For non-workshop positions $j \in \mathcal{J} \setminus \mathcal{W}$, the integrality of x_{ij} enforces the step size directly (integer for tutorials, multiple of 0.5 for seminars and marking).

To link x_{ij} to y_{ij} , we use a big- M constraint where $M = \max_i u_i$. This ensures $x_{ij} = 0$ whenever $y_{ij} = 0$, since the objective minimizes cost and will never assign BUs to an unassigned position:

$$x_{ij} \leq M \cdot y_{ij} \quad \forall i \in \mathcal{A}, j \in \mathcal{J}.$$

Note that a lower bound constraint $x_{ij} \geq s_j \cdot y_{ij}$ is not needed: the position BU needs (Constraint 1) force enough BUs to be assigned overall, and the integrality of x_{ij} already ensures at least one step is assigned whenever $y_{ij} = 1$.

4. **At most one workshop per applicant.**

$$\sum_{j \in \mathcal{W}} y_{ij} \leq 1 \quad \forall i \in \mathcal{A}.$$

5. **At most one tutorial or seminar per applicant.**

$$\sum_{j \in \mathcal{J}_{\text{TS}}} y_{ij} \leq 1 \quad \forall i \in \mathcal{A}.$$

6. **At most three positions per applicant.**

$$\sum_{j \in \mathcal{J}} y_{ij} \leq 3 \quad \forall i \in \mathcal{A}.$$

Model: Complete Formulation.

$$\begin{aligned}
\min \quad & \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i (x_{ij} + p_{ij} y_{ij}) \\
\text{subject to} \quad & b_j^{\min} \leq \sum_{i \in \mathcal{A}} x_{ij} \leq b_j^{\max} & \forall j \in \mathcal{J} \\
& \ell_i \leq \sum_{j \in \mathcal{J}} x_{ij} \leq u_i & \forall i \in \mathcal{A} \\
& x_{ij} \in \mathbb{Z}_{\geq 0} & \forall i \in \mathcal{A}, j \in \mathcal{T} \\
& x_{ij} \in \{0, 0.5, 1, 1.5, \dots\} & \forall i \in \mathcal{A}, j \in \mathcal{M} \cup \mathcal{S} \\
& x_{ij} \leq M y_{ij} & \forall i \in \mathcal{A}, j \in \mathcal{J} \\
& \sum_{j \in \mathcal{W}} y_{ij} \leq 1 & \forall i \in \mathcal{A} \\
& \sum_{j \in \mathcal{J}_{\text{TS}}} y_{ij} \leq 1 & \forall i \in \mathcal{A} \\
& \sum_{j \in \mathcal{J}} y_{ij} \leq 3 & \forall i \in \mathcal{A} \\
& x_{ij} \geq 0 & \forall i \in \mathcal{A}, j \in \mathcal{J} \\
& y_{ij} \in \{0, 1\} & \forall i \in \mathcal{A}, j \in \mathcal{J}.
\end{aligned}$$

This is a Mixed Binary Integer Linear Program (MBILP) with $|\mathcal{A}| \cdot |\mathcal{J}| = 20 \times 9 = 180$ binary variables y_{ij} and up to 180 continuous/integer variables x_{ij} .

Question 2: Augmented Model with Applicant Preferences

Overview. Each applicant $i \in \mathcal{A}$ submits an ordered preference list of up to four positions. To account for preferences in a simple and tractable way, we introduce a binary indicator for each applicant that tracks whether they receive **any of their four preferred positions**. Our goal is to ensure that a minimum fraction of applicants receive at least one preferred appointment, while still minimizing total cost.

This approach is simpler and more transparent than a full penalty-based method: it avoids the need to calibrate a penalty weight λ , and the coverage parameter α has a direct and intuitive interpretation.

New Variable. Introduce a binary indicator $z_i \in \{0, 1\}$ for each applicant $i \in \mathcal{A}$, where $z_i = 1$ if applicant i is assigned to any of their preferred positions, and $z_i = 0$ otherwise.

New Constraints.

1. **Link z_i to preferred assignments.** The variable z_i can equal 1 only if applicant i is actually assigned to at least one of their four preferred positions $j_i^1, j_i^2, j_i^3, j_i^4$:

$$z_i \leq \sum_{k=1}^4 y_{i,j_i^k} \quad \forall i \in \mathcal{A}.$$

2. **Minimum preference coverage.** At least a fraction $\alpha \in [0, 1]$ of all applicants must receive at least one of their preferred appointments:

$$\sum_{i \in \mathcal{A}} z_i \geq \alpha |\mathcal{A}|.$$

For example, setting $\alpha = 0.5$ requires that at least 10 out of 20 applicants receive a preferred position.

3. **Integrality.**

$$z_i \in \{0, 1\} \quad \forall i \in \mathcal{A}.$$

Objective Function. The objective from Question 1 is unchanged:

$$\min \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i (x_{ij} + p_{ij} y_{ij}).$$

Preferences enter only through the new constraints, not the objective. This keeps the cost minimization goal clear and avoids introducing an arbitrary penalty weight.

Complete Augmented Formulation.

$$\begin{aligned}
& \min \quad \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i (x_{ij} + p_{ij} y_{ij}) \\
& \text{subject to} \quad \text{all constraints from Question 1,} \\
& \quad z_i \leq \sum_{k=1}^4 y_{i,j_i^k} \quad \forall i \in \mathcal{A} \\
& \quad \sum_{i \in \mathcal{A}} z_i \geq \alpha |\mathcal{A}| \quad (\alpha = 0.5) \\
& \quad z_i \in \{0, 1\} \quad \forall i \in \mathcal{A}.
\end{aligned}$$

This adds only $|\mathcal{A}| = 20$ binary variables z_i and $|\mathcal{A}| + 1 = 21$ constraints to the model from Question 1, keeping the overall structure as an MBILP.

Discussion. We account for all four submitted preferences rather than just the first choice, which gives the model more flexibility and reduces the risk of infeasibility.

- **Simplicity.** The model adds only one new parameter α whose meaning is immediately clear: the minimum fraction of applicants who receive at least one preferred appointment.
- **No calibration needed.** A penalty-based approach would require choosing a weight λ to balance monetary cost against preference satisfaction, with no natural scale. Our constraint-based approach avoids this entirely.
- **Feasibility control.** If α is set too high the model may become infeasible — for example, if many applicants share the same preferred positions but those positions can only absorb a limited number of TAs. Considering all 4 choices significantly reduces this risk compared to only tracking first choices. Lowering α relaxes the constraint until a feasible solution exists.
- **Choice of α .** A reasonable starting value is $\alpha = 0.5$, meaning at least half of applicants receive one of their preferred positions. Higher values of α prioritize preference satisfaction at the potential cost of a higher total assignment cost, while lower values prioritize cost minimization. If the model is infeasible at $\alpha = 0.5$, one can lower α gradually until feasibility is restored.

Question 3: Implementation

Overview. Implementing the MBILP from Question 1 proved challenging due to the scale of the model. We attempted two approaches before arriving at a working solution. We describe all three approaches below for completeness.

Approach 1: Excel Solver (attempted, unsuccessful). Our first attempt was to implement the full model directly in Excel Solver, as suggested in the project instructions. However, the full model has approximately 360 decision variables (180 binary y_{ij} and up to 180 continuous/integer x_{ij}), which exceeds Excel Solver’s hard limit of 200 decision variables. Even after simplifying by dropping the n_{ij} auxiliary variables and relying directly on integrality of x_{ij} , the variable count remained too large for Excel Solver to accept. This approach was therefore abandoned.

Approach 2: Python/CBC solver (considered, not adopted as primary). As an alternative, we implemented the full model in Python using the PuLP modelling library with the open-source CBC solver. This produced a globally optimal solution with a total cost of \$106,671.26 (466 branch-and-bound nodes, approximately 5 seconds). However, since the project hint specifically suggested LibreOffice or manual decomposition as alternatives, and since Python is not a standard tool in this course, we decided to also develop a fully Excel-based approach that better fits the spirit of the assignment. The Python solution is kept as a backup for verification.

Approach 3: Two-step decomposition in Excel (adopted). Following the project hint to “*decide a limited number of appointments by hand in order to reduce the number of decision variables,*” we split the problem into two stages:

1. **Step 1 — Manual workshop assignment.** We assigned applicants to the two workshops (MATH-ACW and MATHCW) by hand, based on their stated preferences. Since workshops have no step-size restrictions and the applicant preference lists make the assignment natural, this step requires no solver. We assigned:
 - **MATHACW** (needs 20–25 BUs): App2 (3 BU, 1st choice), App5 (3 BU, 1st choice), App7 (3 BU, 1st choice), App11 (4 BU, 4th choice), App13 (2 BU, 2nd choice), App17 (3 BU, 3rd choice), App18 (3 BU, 1st choice). Total: 21 BUs $\in [20, 25]$. ✓
 - **MATHCW** (needs 12–16 BUs): App8 (2 BU, 3rd choice), App9 (2 BU, 1st choice), App14 (3 BU, 1st choice), App19 (4 BU, 1st choice), App20 (2 BU, 4th choice). Total: 13 BUs $\in [12, 16]$. ✓

Every applicant assigned to a workshop listed that workshop among their four preferences, so this manual step also satisfies the spirit of Question 2.

2. **Step 2 — Excel Solver for remaining positions.** With 12 applicants fixed to workshops, only 8 applicants and 7 positions remain: MATH150, MATH254, MACM316, MATH260, MATH338, MATH342, and MATH426. This gives $8 \times 7 = 56$ decision variables x_{ij} — well within Excel Solver’s limit of 200. We set up the model in Excel with the objective cell equal to the total cost of the remaining positions, and added the following constraints:

- Course BU bounds: $b_j^{\min} \leq \sum_i x_{ij} \leq b_j^{\max}$ for each remaining position.

- Applicant BU bounds: $\ell_i \leq \sum_j x_{ij} \leq u_i$ for each remaining applicant.
- Step size: $x_{ij} \in \mathbb{Z}_{\geq 0}$ for tutorial positions; x_{ij} a non-negative multiple of 0.5 for seminar and marking positions.
- At most one tutorial or seminar per applicant.
- Non-negativity: $x_{ij} \geq 0$.

The Excel Solver found an optimal solution for this subproblem.

Results. The combined assignment is shown in Table 1. The total cost is:

$$\underbrace{\$61,542.98}_{\text{workshops (manual)}} + \underbrace{\$49,101.48}_{\text{remaining (solver)}} = \$110,644.46.$$

Applicant	Prog.	M150	M254	M316	M260	M338	M342	M426	ACW	CW
App1	PhD	1.5	—	—	—	—	—	—	—	—
App2	MSc	—	—	—	—	—	—	—	3.0	—
App3	PhD	—	—	—	—	0.5	—	0.5	—	—
App4	MSc	—	—	5.0	—	—	—	—	—	—
App5	PhD	—	—	—	—	—	—	—	3.0	—
App6	MSc	—	3.0	—	—	—	—	—	—	—
App7	MSc	—	—	—	—	—	—	—	3.0	—
App8	PhD	—	—	—	—	—	—	—	—	2.0
App9	PhD	—	—	—	—	—	—	—	—	2.0
App10	PhD	—	—	—	—	—	3.0	—	—	—
App11	MSc	—	—	—	—	—	—	—	4.0	—
App12	PhD	—	—	—	4.0	—	—	—	—	—
App13	PhD	—	—	—	—	—	—	—	2.0	—
App14	MSc	—	—	—	—	—	—	—	—	3.0
App15	PhD	—	—	—	2.0	—	—	—	—	—
App16	MSc	—	—	4.0	—	—	—	—	—	—
App17	MSc	—	—	—	—	—	—	—	3.0	—
App18	PhD	—	—	—	—	—	—	—	3.0	—
App19	MSc	—	—	—	—	—	—	—	—	4.0
App20	MSc	—	—	—	—	—	—	—	—	2.0
Total		1.5	3.0	9.0	6.0	0.5	3.0	0.5	20.0	13.0
$b_j^{\min}$		1.5	3.0	9.0	6.0	0.5	3.0	0.5	20.0	12.0
$b_j^{\max}$		2.0	3.0	9.0	7.0	1.5	3.0	1.5	25.0	16.0

Table 1: Final TA assignment (non-prep BUs). All course totals satisfy $b_j^{\min} \leq \sum_i x_{ij} \leq b_j^{\max}$.

Analysis of the solution.

- **All constraints are satisfied.** Every course receives BUs within its required range, every applicant's total worked BUs fall within $[\ell_i, u_i]$, step sizes are respected, no applicant is assigned to two workshops or two tutorials/seminars, and no applicant holds more than three positions.
- **Preference alignment.** All 12 workshop-assigned applicants listed their assigned workshop in their top-4 preferences. This naturally satisfies the spirit of the augmented model from Question 2.

- **Cost comparison.** The two-step solution costs \$110,644.46, compared to \$106,671.26 from the Python/CBC global optimum. The difference of approximately \$3,973 (about 3.7%) reflects the sub-optimality introduced by fixing workshop assignments manually rather than optimizing them jointly with the remaining positions. This is a reasonable trade-off given the constraint imposed by Excel Solver's variable limit.
- **Cost minimization behaviour.** Both workshops are assigned at or near their minimum required BU level (MATHACW: 21 BU, MATHCW: 13 BU), since additional BUs beyond the minimum would increase cost. Most applicants receive exactly one position, minimizing prep BU overhead since prep BUs are paid once per position assigned.
- **App3** is assigned two marking positions (MATH338 and MATH426, 0.5 BU each), totalling 1 BU and satisfying their fixed range $[1, 1]$. This is permitted since the constraint only restricts applicants to at most one tutorial or seminar.

Note on Preferences. Applicant preferences from Question 2 are not formally incorporated into the objective or constraints of this model beyond the manual workshop step. Fully incorporating preferences would require the augmented model from Question 2, adding binary indicator variables z_i and a coverage constraint $\sum_{i \in \mathcal{A}} z_i \geq \alpha |\mathcal{A}|$.